

Deepak Mallampati

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F-1 OPT - sponsorship welcome

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping agentic systems for regulated enterprise workflows - full-stack from retrieval pipelines to deployed APIs, with direct customer-facing delivery ownership. Built multi-agent compliance pipelines, production RAG with ~35% retrieval precision gain, and LLM workflows with >30% hallucination reduction. Comfortable owning the full technical relationship with enterprise customers: architecting solutions, translating requirements, and iterating from feedback. Stack: Python, TypeScript, LangChain, LlamaIndex, FastAPI, Docker.

CORE COMPETENCIES

Agentic LLM Systems • Multi-Agent Orchestration • Retrieval-Augmented Generation (RAG) • Evaluation Pipelines • Full-Stack Delivery (Python • TypeScript • React) • FastAPI / Flask / Docker • Prompt & Context Engineering • LangChain • LlamaIndex • Vector Search & Embeddings • HIPAA-Conscious AI Governance • Customer-Facing Technical Delivery • CI/CD • MLOps • Observability

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology & AI Consulting
USA (Remote)

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Built **RAG systems** for clinical knowledge retrieval using recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and raised retrieval-grounded response alignment above 90% in evaluation.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%; translated enterprise feedback into pipeline improvements.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by 15-20% on high-risk categories.
- Maintained HIPAA-conscious AI governance: de-identification, evaluation audit trails, and stakeholder-facing system-limitation documentation for regulated enterprise delivery.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer - Oil & Gas Enterprise ERP
Hyderabad, India

- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed Grafana dashboards tracking long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence ~25%.
- Optimized T-SQL stored procedures powering Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Video Analytics & NLP

- Built transformer-based video summarization system; reduced manual review time 60-70% (hours → <5 minutes per video) and achieved ~85% highlight alignment.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30%; deployed as Flask API for non-technical staff.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN-based duplicate claim detection; explainable risk scoring with audit-ready reasoning traces - compliance architecture identical to enterprise legal workflow automation.

Python · LangChain · LlamaIndex · FastAPI · Docker · Vector Search

Manga Lens shipped · Chrome Web Store

Full-stack Chrome extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-section capture pipeline, 7-day translation cache, per-domain selector configs for 29 sites - demonstrates multi-provider agentic integration and real-world API orchestration at scale.

TypeScript · Manifest V3 · Service Workers · Claude API · GPT-4o

Dream Decoder - Multimodal AI Platform

Full-stack FastAPI + React/TypeScript/Vite app orchestrating coordinated multimodal APIs (Perplexity Sonar + Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%**.

Python · FastAPI · React · TypeScript · Vite · Tailwind · Perplexity · Replicate

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced two-stage CNN with unified YOLOv8; reduced inference latency ~**30%**; sliding-window confidence aggregation suppressed blink-driven false positives ~**25%**; adaptive frame skipping for real-time operation.

Python · YOLOv8 · OpenCV · PyTorch

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL (T-SQL, PostgreSQL), React, PHP

LLM & Agentic: LangChain, LlamaIndex, agentic workflows, RAG, structured outputs, prompt engineering, evaluation pipelines, grounding, vector search

ML: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers

CV & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Infra: FastAPI, Flask, Docker, Jenkins, Grafana, CI/CD, RESTful APIs, MLOps/LLMOps, observability

Domains: Healthcare AI (HIPAA-conscious), Enterprise AI, Agentic Compliance Systems, Workflow Automation